

AI Photographer

PRD + Implementation Plan

1. Problem

Professional photographer: \$200-500+, weeks to schedule. Friend with your phone: badly framed, badly lit, badly timed. The gap is the *eye*.

2. Target User

- **Subject(s)**: 1 person, couple, friends, family -- any group. Already dressed up.
- **Camera-holder**: friend/family, no photography knowledge, follows on-screen guidance.

Out of scope v1: solo/selfie/tripod mode.

3. The Style

ONE signature photographic style. The 3 reference photos define a single aesthetic DNA:

Golden Hour Editorial

Light -- dramatic, directional, warm. Strong shadow/highlight contrast. Never flat. Sidelight for depth, backlight for glow, found light patterns for texture on warm surfaces.

Subject-environment -- environment-forward. Subjects share frame with surroundings or are deliberately secondary (small, blurred, or underexposed). Environment tells PLACE; subjects tell EMOTION.

Emotion -- authentic, in-motion, unposed. Walking, laughing, whispering, embracing. Never stiff. Direct behavior ("walk slowly"), not poses. Best frame is mid-action.

Color -- warm amber/gold. Shadows crushed near-black. Subtle vignette. "Golden hour even when it isn't."

Composition -- environmental framing. Archways, corridors, tree canopies, light patterns, leading lines. Subjects placed within the environment's structure.

Technique -- scene-adaptive. Motion blur for energy, scale framing for place-making, exposure play for drama, or clean capture when light does the work.

4. User Flow

1. Open app -> see the style with reference images
2. Register subjects: quick photo of everyone who'll be in the shots
3. Scene analysis (two paths):
 - a. SCOUT: video surroundings -> app finds optimal spot -> guides you there
 - b. DIRECT: point at a scene -> app evaluates it
4. App generates coaching for this group + this scene
5. Director's view: live cues (framing, brightness, position) as overlays + audio
6. Auto-readiness: real-time video detects when subjects follow cues -> app automatically adjusts camera settings (exposure, focus, WB)
7. Camera-holder captures photo
8. App shows best frame / edited preview
9. User saves selected photos
10. App logs keeper signal locally

Step Details

Step 1 -- Style Introduction:

User opens the app and sees the "Golden Hour Editorial" style presented with reference images. A brief description of the aesthetic (warm, dramatic light, environment-forward, candid emotion) sets expectations. Camera-holder understands the look before shooting. Single "Start Shooting" button proceeds to subject registration.

Step 2 -- Subject Registration:

Camera-holder takes a quick photo of all subjects (can be a selfie if camera-holder is also a subject, or a group shot). This gives the app:

- Subject count (auto-detected via VNDetectHumanRectanglesRequest)
- Visual reference of the group (sent to vision model with the scene for context-aware coaching)
- Approximate group composition (adults/kids can be inferred from relative heights)

Step 3a -- Scout Mode:

User holds phone up and slowly pans surroundings. App runs on-device scene quality checks (brightness variance, leading line detection via edge detection). When on-device checks look promising:

- App signals "This looks interesting -- hold still"
- Sends frame to vision model for style-fit evaluation
- Model returns a score + reason ("Great directional light and architectural depth -- 8/10 for this style")
- When good enough: "This is a great spot. Stay here." + saves a reference screenshot

Step 3b -- Direct Mode:

User points at a chosen scene. App sends frame to model for evaluation and shot design in one call.

Step 4 -- Coaching Generation:

After scene analysis, the app generates a shot plan tailored to this specific group in this specific scene. The vision model returns: which capture technique to use, where the camera-holder should stand, step-by-step subject directions (e.g., "Walk slowly toward the arch," "Whisper something to each other"), how to use the available light, the key moment to capture, and any grooming fixes. This coaching is displayed as a card and spoken aloud before entering Director's View.

Step 5 -- Live Cues:

Short, clear on-screen text + arrows. Based on real-time on-device analysis:

- Framing: "Back up" / "Move subjects left" / correction arrows
- Brightness: light direction indicator
- Position: subject zone overlay (white -> orange -> green)
- Grooming: "Push bangs aside" / "Relax shoulders"

Step 6 -- Auto Camera Settings:

Once subjects are positioned (readiness checks green), app automatically:

- Locks focus on subjects
- Adjusts exposure for the scene's style technique
- For exposure_play: auto-detects brightest region, locks exposure there, applies bias -- NO manual tap required
- Sets white balance toward warm
- Camera-holder just needs to press the shutter at the right moment

5. User Stories

#	As a...	I want to...	So that...
---	---------	--------------	------------

U1	Camera-holder	Register who's in the photo before shooting	The app gives coaching specific to my group
U2	Camera-holder	Pan around and have the app find the best spot	I don't need a photographer's eye for location scouting
U3	Camera-holder	See live on-screen cues for positioning	I can direct subjects visually
U4	Camera-holder	Have camera settings adjusted automatically	I never think about exposure or focus
U5	Subject	Hear what to do without seeing the screen	I know when to walk/hold/laugh
U6	Camera-holder	Take multiple photos and pick the best	Only good photos go to camera roll
U7	Camera-holder	Get grooming reminders (bangs, posture)	Small details don't ruin the shot

6. Functional Requirements

6.1 Style System

FR-1: One style ("Golden Hour Editorial") defined as a structured prompt.

FR-2: 4 capture techniques as tools (model-selected per scene).

FR-3: For beta, one style. Architecture supports adding more styles later.

6.2 Subject Registration

FR-4: Camera-holder photographs subjects before shooting. App auto-detects count.

FR-5: Subject photo sent to vision model alongside scene for context-aware coaching.

6.3 Scene Analysis

FR-6: Scout mode: on-device quality checks during pan, model call on promising frames.

FR-7: Direct mode: single model call with scene frame + subject photo + style.

FR-8: Model returns: technique, camera position, subject directions, light usage, grooming.

FR-9: If scene doesn't suit the style, model explains why and suggests what to look for.

6.4 Director's View + Auto-Readiness

FR-10: On-device real-time overlays: subject zone, correction arrows, grooming card.

FR-11: When subjects follow cues and readiness checks are green, app auto-adjusts camera settings for the selected technique.

FR-12: For exposure_play: app auto-detects brightest region and locks exposure there. No manual tap.

FR-13: Camera-holder presses shutter when ready. Settings are already optimal.

6.5 Capture + Review

FR-14: 4 capture techniques implemented as tools (model-selected).

FR-15: Best frame shown with finishing applied. Before/after toggle.

FR-16: Multi-photo: session strip, take another, review gallery, keep/delete, batch save.

6.6 Analytics

FR-17: Each save logs style ID + technique + keeper signal locally.

7. Success Metrics

Metric	Definition	Target	Source
Keep rate	saved / captured per session	>=40%	Session data
Session yield	Photos saved per session	>=3	Session data
Retake ratio	captured / saved	<4x	Session data
Session completion	Sessions with >=1 save	>=70%	Session data
Crash-free	Sessions without crash	>=95%	Xcode Organizer

8. Model Strategy

Two-layer architecture

On-device (Apple Vision) -- real-time, private, free:

- VNDetectHumanBodyPoseRequest -- positioning + grooming
- VNDetectFaceLandmarksRequest -- grooming (bangs, tilt, expression)
- VNDetectFaceRectanglesRequest -- backs-to-camera
- VNDetectHumanRectanglesRequest -- bounding box, subject count, correction vector
- Brightness/edge analysis -- scout mode on-device pre-filter

Cloud vision model -- 1-2 calls per session:

- Call 1: Scene + subjects + style -> shot plan (technique + coaching)
- Call 2 (optional): Post-capture -> feedback

Model: cheapest effective multimodal, pluggable.

Default: Gemini 2.0 Flash (~\$0.10/1M tokens, ~1-2s). Pluggable provider protocol (swap to GPT-4o-mini, Claude Haiku, etc.).

API key: Secrets.plist (gitignored) in binary for TestFlight. Server proxy for App Store (out of scope v1).

Privacy: Only the scan frame (1280x960, 180KB) and subject registration photo leave the device. Final photos stay local. "Your scene is briefly analyzed to design the shot. Photos are not stored."

9. Scope

In v1

- One style, scene-adaptive
- Subject registration
- Scene analysis: scout mode + direct mode
- 4 capture techniques (model-selected)
- Auto camera settings on readiness
- Director's view with live cues
- Multi-photo session with gallery
- CoreImage finishing
- Local keeper analytics
- TestFlight

Not in v1

- Multiple styles / style marketplace
- Solo/tripod mode
- Server-side API proxy
- User accounts / cloud sync

Implementation Plan

Vision Model Prompts

Prompt 1 -- Shot Design (main call)

Sent with: scene frame (1280x960) + subject registration photo + group description.

You are a professional photographer with this signature style:

LIGHT: Dramatic, directional, warm. Golden hour or warm-toned natural light. Strong shadow/highlight contrast -- never flat. Shadows kept deep. Sidelight for depth, backlight for glow, found light patterns for texture.

SUBJECT-ENVIRONMENT: Environment-forward. Subjects share frame with surroundings or are deliberately secondary: small (scale), in motion (blur), or underexposed (drama). Environment = PLACE. Subjects = EMOTION.

EMOTION: Authentic, in-motion, unposed. Walking, laughing, whispering, embracing. Never stiff, rarely at camera. Direct behavior, not poses. Best frame = mid-action.

COLOR: Warm amber/gold. Crushed shadows. Vignette. "Golden hour even when it isn't."

COMPOSITION: Environmental framing -- archways, corridors, trees, light patterns. Subjects within the environment's structure.

CAPTURE TECHNIQUES (pick one based on the scene):

- "motion_blur": 2-second motion capture. Best for paths/corridors with directional light.
- "scale_framing": subjects very small ($\leq 20\%$) against grand backdrop.
- "exposure_play": expose for bright surface, underexpose subjects. Best for light patterns on warm surfaces.
- "clean_capture": sharp standard shot when light + composition are strong.

SUBJECTS: [attached subject photo] -- {group_description}

SCENE: [attached scene photo]

Design a shot. If the scene doesn't suit this style, say so and suggest what to look for nearby.

Reply JSON:

```
{
  "scene_works": true/false,
  "scene_issue": "...", // only if false -- what's wrong + what to look for
  "capture_technique": "motion_blur" | "scale_framing" | "exposure_play" | "clean_capture",
  "camera_position": "where to stand, angle, height",
  "subject_directions": ["direction 1", "direction 2", "direction 3"],
  "light_usage": "how to use the light in this specific scene",
  "key_moment": "what action/gesture to capture",
  "grooming_check": "quick fix before shooting"
}
```

Prompt 1b -- Scout Mode Scene Rating

Sent with: promising frame during panning.

Rate this scene for the "Golden Hour Editorial" style (warm, dramatic light, environment-forward, candid emotion). Score 1-10 with one-sentence reason.
Reply JSON: { "score": N, "reason": "..."} }

Prompt 2 -- Post-Capture Feedback (optional)

This photo was shot in "Golden Hour Editorial" style (warm, environment-forward, candid, crushed shadows). Group: {group_description}.
ONE specific improvement. 15 words max.

Auto Camera Settings (per technique)

When readiness checks go green, app auto-adjusts without camera-holder input:

Technique	Auto-Adjustment
motion_blur	`preferredVideoStabilizationMode = .off`; continuous autofocus locked; exposure set for the scene
scale_framing	Standard AF/AE; focus on landmark region; ensure deep depth of field
exposure_play	Auto-detect brightest region -> `setExposurePointOfInterest` -> settle 800ms -> lock -> bias -1.5 -> focus lock on subjects
clean_capture	Standard AF/AE; focus on nearest face; flash off

For exposure_play auto-detection: FrameAnalyzer computes the brightest region by dividing the frame into a grid (e.g., 6x6), computing mean luminance per cell, and selecting the cell with highest value. The exposure point is set to that cell's center -- no manual tap needed.

Director's View

On-Device Overlays (Apple Vision at 5fps)

Subject Zone -- technique-specific position and size:

- motion_blur: center-bottom, subjects walk away
- scale_framing: bottom 20%, subjects small
- exposure_play: center, in front of bright surface
- clean_capture: center thirds

Color: white -> orange -> green.

Correction Arrows -- from correctionVector. Dead zone 0.05.

Readiness Checklist -- per technique:

Check	motion	scale	exposure	clean
Subject detected	Y	Y	Y	Y
Subject in zone	Y	Y	Y	Y
Backs to camera	Y	--	--	--
Subject <=20%	--	Y (%)	--	--

All green -> auto camera adjustment -> capture button pulses.

Grooming Card -- on-device:

Detection	Cue
Eye landmarks occluded	"Push bangs out of eyes"
Shoulders uneven/raised	"Relax shoulders"
Head tilt >15 deg	"Straighten head"
Stiff expression	"Say something natural"
Arms rigid	"Relax arms"
Generic	"Straighten clothes"

Model-Generated Content**Coaching Card -- from Prompt 1:**

- subject_directions as numbered list
- light_usage callout
- key_moment before capture
- Spoken aloud by AudioCoach

Data Flow

STEP 2 -- SUBJECT REGISTRATION:

Camera -> photo -> VNDetectHumanRectanglesRequest (count) -> store subjectPhoto

STEP 3 -- SCENE ANALYSIS:

Scout mode: camera pan -> FrameAnalyzer (brightness variance, edges) ->

promising frame -> Prompt 1b (score) -> "good spot!" or keep panning

Direct mode: camera -> snapshot -> Prompt 1 (scene + subjects + style) -> ShotPlan

STEP 5-6 -- DIRECTOR'S VIEW:

AVCaptureVideoDataOutput (30fps) -> FrameAnalyzer (5fps) ->

FrameAnalysis -> DirectorOverlay (overlays + grooming)

-> readiness checks -> when all green:

auto camera settings (per technique)

capture button activates

STEP 7 -- CAPTURE:

CaptureEngine (technique from ShotPlan) -> CIImage ->

PhotoFinisher (CoreImage) -> session.photos.append()

STEP 8-9 -- REVIEW + SAVE:

SessionGallery -> keep/delete -> batch PHPhotoLibrary save -> keeper tracking

File Structure

```

AIPhotographer/
+-- App/
|   +-- AIPhotographerApp.swift
+-- Models/
|   +-- Style.swift           # ONE style definition
|   +-- ShotPlan.swift        # Model output: technique + coaching
|   +-- Session.swift         # Shoot state + isReadyToCapture
|   +-- PhotoSession.swift    # Multi-photo session + CapturedPhoto

```

```

+-- Camera/
|   +-- CameraManager.swift      # AVFoundation, dual output, auto-settings
|   +-- CaptureEngine.swift      # 4 capture techniques
|   +-- FrameAnalyzer.swift      # Apple Vision at 5fps, grooming, scout quality
|   +-- AutoSettings.swift       # auto exposure/focus/WB per technique
+-- VisionModel/
|   +-- VisionModelProvider.swift # Protocol + pluggable providers
|   +-- ShotDesigner.swift        # Prompt 1 (shot plan), 1b (scout), 2 (feedback)
+-- Coaching/
|   +-- AudioCoach.swift         # AVSpeechSynthesizer
|   +-- DirectorOverlay.swift    # On-device overlays
|   +-- CoachingCardView.swift   # Model coaching display
+-- Finishing/
|   +-- PhotoFinisher.swift      # CoreImage chain, per-technique params
+-- Analytics/
|   +-- SessionTracker.swift      # Keeper metrics
+-- Views/
|   +-- StyleIntroView.swift      # Style + reference images (landing screen)
|   +-- SubjectRegView.swift      # photograph the subjects
|   +-- SceneAnalysisView.swift   # scout mode + direct mode
|   +-- SetupView.swift          # Director's view with overlays
|   +-- CaptureView.swift        # Capture + session strip
|   +-- SessionGalleryView.swift  # Keep/delete, batch save
|   +-- ReviewView.swift         # Full-screen, before/after, feedback

```

3-Day Build Sequence

Day 1 -- Foundation (8 hours)

- **Hour 1:** Xcode project, capabilities, Info.plist, TestFlight provisioning
- **Hours 2-3:** Style.swift, ShotPlan.swift, Session.swift, PhotoSession.swift
- **Hours 3-5:** CameraManager.swift -- dual output, sessionQueue, UIViewRepresentable, portrait lock
- **Hours 5-6:** VisionModelProvider.swift + ShotDesigner.swift -- provider protocol, Gemini Flash impl, Prompts 1/1b/2, JSON parsing
- **Hours 6-8:** StyleIntroView.swift + SubjectRegView.swift + SceneAnalysisView.swift -- landing, subject photo capture, scout/direct mode UI

Day 2 -- Intelligence (8 hours)

- **Hours 1-3:** FrameAnalyzer.swift -- Apple Vision (pose, face landmarks, face rects, human rects), 5fps throttle, grooming, brightness analysis for scout mode, correctionVector
- **Hours 3-4:** AutoSettings.swift -- per-technique auto exposure/focus/WB; auto brightest-region detection for exposure_play
- **Hours 4-6:** DirectorOverlay.swift + CoachingCardView.swift -- technique-specific overlays, correction arrows, readiness checklist, grooming card, coaching display
- **Hours 6-8:** CaptureEngine.swift + AudioCoach.swift -- 4 capture techniques + audio synthesis + SetupView + CaptureView wiring

Day 3 -- Polish + Ship (6 hours)

- **Hours 1-2:** PhotoFinisher.swift -- CoreImage chain, per-technique finishing params
- **Hours 2-3:** SessionGalleryView.swift + ReviewView.swift + SessionTracker.swift -- gallery, keep/delete, batch save, before/after, feedback, analytics
- **Hours 3-4:** End-to-end device test at 3+ locations

- **Hours 4-5:** Archive -> TestFlight -> testers
- **Hour 6:** Bug buffer

Known Risks

Risk	Likelihood	Mitigation
Model selects wrong technique	Medium	Coaching card shown first; camera-holder confirms or rescans
Model coaching too generic for specific scene	Medium	Prompt includes both scene AND subject photos; detailed style definition
Scout mode model calls too frequent/costly during pan	Medium	On-device pre-filter (brightness variance) gates model calls; max 1 call per 5s
Auto exposure_play locks on wrong bright region	Medium	Grid-based brightest-region detection; if result looks wrong, camera-holder can manually override
Model says scene doesn't work too often	Medium	Prompt wording emphasizes finding what DOES work; suggest nearby alternatives
4+ Vision requests at 5fps -> thermal	Medium	Profile Day 2; reduce to 3fps if needed

Verification

1. **Subject registration:** Photo 1, 2, and 4+ people. Verify auto-count. Verify subject photo appears in model prompt.
2. Scout mode: Pan across 3+ locations. Verify on-device quality check triggers model call on promising frames. Verify "stop here" signal.
3. Direct mode: Point at 5 diverse scenes (not reference locations). Verify model returns sensible technique + coaching.
4. Auto settings: Verify exposure_play auto-detects brightest region without manual tap. Verify settings lock after readiness green.
5. Each technique: motion blur composite, scale framing overlay, exposure play, clean capture -- all produce usable photos.
6. **Group variety:** Solo, 2 friends, family of 4. Coaching differs meaningfully.
7. **Multi-photo:** 5 captures, keep 3, batch save -> 3 in camera roll.
8. **Style consistency:** Compare outputs to reference photos. Do they share the aesthetic DNA?
9. **Analytics:** SessionTracker logs keep_rate, session_yield, technique used.